Engineering Spin States in Silicon Quantum Dot-Organic Hybrids

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Introduction

How do we build long-lived, optically addressable spin states for quantum information science?

Quantum information science demands spin states that are both **controllable** and **long-lived** — ideally without the need for cryogenic cooling.

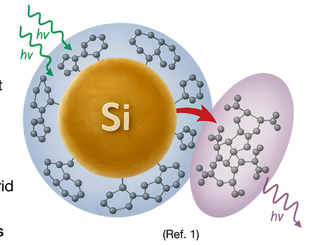
We develop **hybrid systems** combining **silicon quantum dots (Si QDs)** and **π-conjugated organic ligands**, which support spin-polarized excited states.

Why Silicon Quantum Dots?

- Non-toxic
- Long spin lifetimes
- Surface chemistry enables covalent bonding with ligands

Why Organic Molecules?

- Preserve spin coherence → **Weak spin–orbit coupling**
- Ideal for pairing with Si QDs in hybrid spin systems



(Ref. 1)

Spin Qubits: Triplets & Radical Pairs

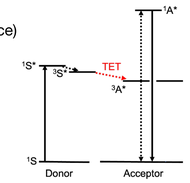
Triplet States (T_1)

- Long-lived, magnetic, and dark → Ideal for spin memory

Spin-Correlated Radical Pairs (SCRPs)

- Entangled spins formed via **charge separation**
- Enable dynamic spin control

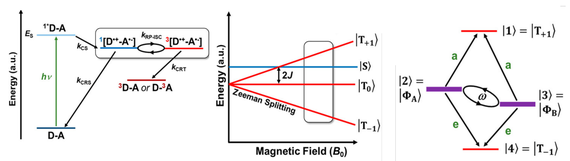
Detection Tool: EPR (Electron Paramagnetic Resonance)



How We Generate Spin States

1. **Triplet Energy Transfer (TET)**
   - Donor → Acceptor via Dexter mechanism
   - Preserves spin polarization — key to coherence
2. **Charge Recombination (CR)**
   - Recombination of electron and hole → triplet state formation
   - Often formed via SCRPs in donor–acceptor pairs

Non-Boltzmann spin polarization → Observable via Transient EPR (TR-EPR)



(Ref. 2)

Acknowledgements

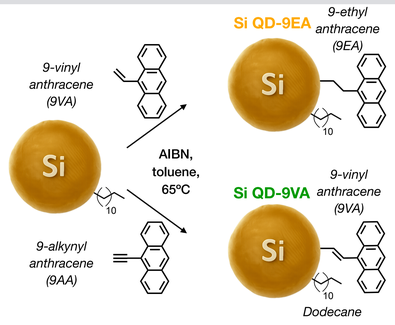


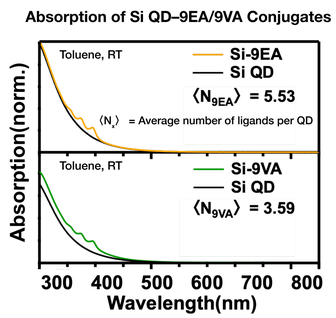
- Argonne National Laboratory
- Amherst College
- NSF award CHE-2147792

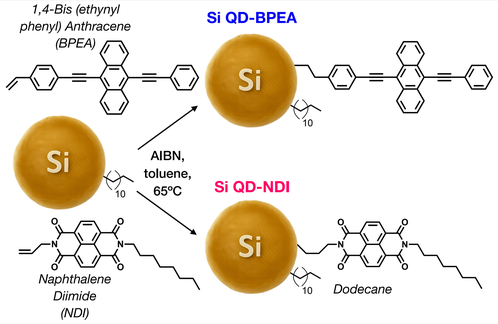
References

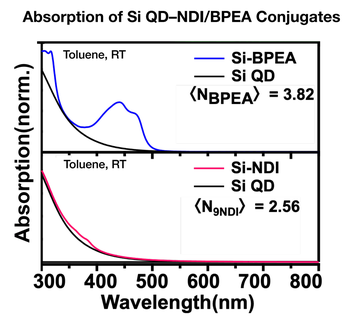
1. Wang, K., and Tang, M.L. *et al*, *Nature Chemistry*, **2023**, 15(8).
2. Harvey, S.M. and Wasielewski, M.R., *JACS*, **2021**, *143*(38).
3. Nguyen, N.Q., and Tang, M.L., *et al*, *Nano Letters*, **2025**, *25*(13).

How Many Molecules per QD? Surface Ligand Quantification of Si QD Conjugates

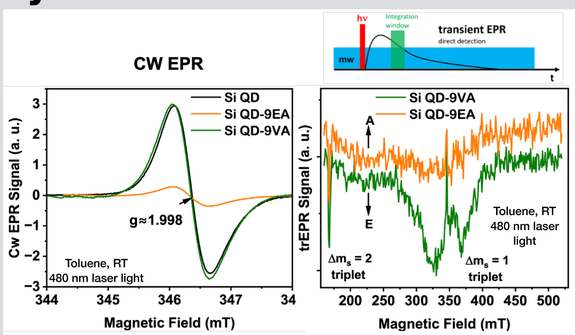








Triplet Transfer in Si QD Hybrids: 9VA Beats 9EA



CW EPR (Left Panel)

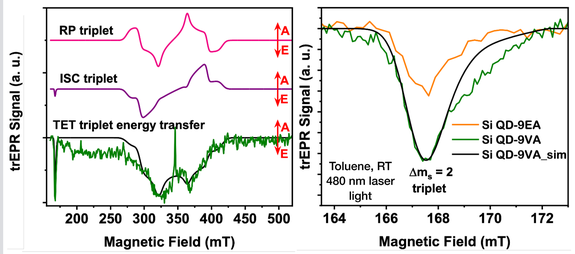
- g ≈ 1.998 narrow line for all samples (Si QD, Si QD–9EA, Si QD–9VA)
- No hyperfine → delocalized conduction-band electrons
- Linewidths (~0.6 mT) nearly identical → similar electron environments
- g-value < defect range (2.002–2.009) → confirms QD origin

TR-EPR (Right Panel)

- Collected 600–1200 ns post-480 nm excitation
- Spin-polarized triplet signals (non-Boltzmann population)

Key transitions:

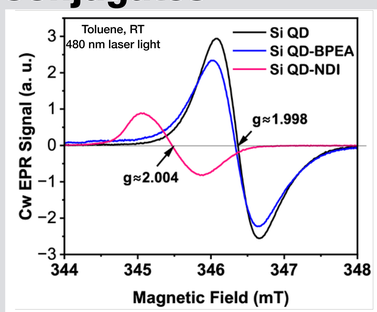
- Δms = 1 → Full-field
- Δms = 2 → Half-field → triplet signature

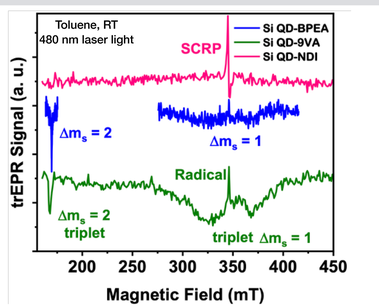


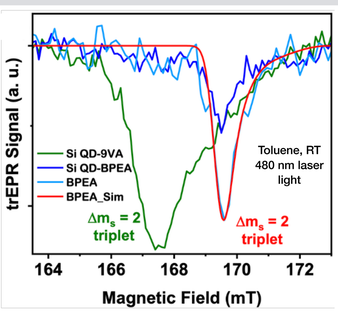
Enhanced Triplet Transfer in Si QD–9VA

- Simulated spectra: 9VA matches TET, not RP or ISC
- Half-field signals match anthracene triplets → ligand-localized
- 9VA shows **~2× stronger** signal than 9EA
- Indicates more efficient TET via enhanced **π-conjugation**

Disentangling Radical vs. Triplet Pathways in Si QD-NDI/BPEA Conjugates







CW EPR Results

- Si QD and Si QD–BPEA show a narrow, delocalized signal at g ≈ 1.998 (linewidth ~0.6 mT).
- Si QD–NDI shows a broader signal at g ≈ 2.004, indicating a **localized radical** on the NDI ligand.

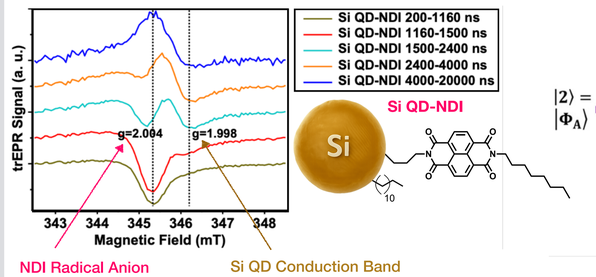
TR-EPR Results

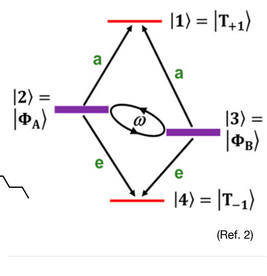
- Si QD–NDI: No triplet signal, but a sharp radical signal → **SCRP signature.**
- Si QD–BPEA: Displays Δms = 2 **half-field signal**, consistent with a triplet state.

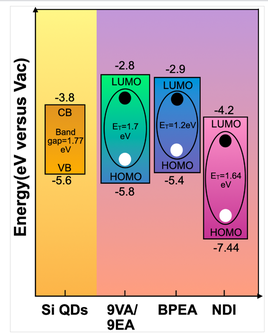
Low-Field (Δms = 2) Comparison

- BPEA and Si QD–BPEA show matching half-field signals → **triplet localized on BPEA.**
- **ZFS position** differs from 9VA/9EA, confirming triplet origin and delocalization pattern.

Spin-Correlated Radical Pair Detection in Si QD–NDI With TR-EPR







- TR-EPR at different integration windows after photo excitation with 480 nm laser light.
- At early times (200–1500 ns), a sharp signal at g = 2.004 indicates the **NDI radical anion.**
- A smaller peak at g = 1.998 matches the **Si QD conduction band.**
- Signal polarity inversion and decay over time reveal **relaxation and dephasing.**
- These features are characteristic of a **spin-correlated radical pair** from:
  → $Si QD^+ + NDI^-$
  → Formed via **photo induced charge separation.**
- Suggests the triplet originates on Si QD and charge transfers to NDI.

Future Work:

- Increase sample concentration to improve triplet signal visibility
- Increase ligand loading on current Si QD size
- Vary QD size to study spin coupling effects
- Use transient absorption to probe excited-state dynamics

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